LABORATORY · Lab · 26 hours

Experiment 3: Time Domain Specifications of a Second-Order System



Control Systems Laboratory (22EEG53)

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Learning objectives

- Determine tr, tp, Mp and ts from the simulated step response
- Compare them with the theoretical values
- Use stepinfo to extract specifications

Syllabus context

Laboratory component of 22EEG53 Control Systems (0:0:2 hours per week). CIE: record and observation 25, viva-voce 5, lab test 20 marks.

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Aim and requirements

Aim: to analyse the time-domain specifications of a second-order system using MATLAB.

Software: MATLAB with the Control System Toolbox (Simulink where required).

Hardware: Personal computer.

System: $C/R = 25/(s^2 + 6s + 25)$: $\omega n = 5$, $\zeta = 0.6$.

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Theoretical formulas

$$t_r = (\pi - \varphi)/\omega_d$$

$\varphi = \cos^{-1}\zeta$

$$t_p = \pi/\omega_d$$

$\omega_d = \omega_n\sqrt{(1-\zeta^2)}$

$$M_p = e^{(-\pi\zeta/\sqrt{(1-\zeta^2)})} \times 100$$

overshoot

$$t_s = 4/(\zeta\omega_n)$$

2 % band

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Procedure

1

Create system

sys = tf(25,[1 6 25]).

2

Simulate

Plot step(sys).

3

Extract

S = stepinfo(sys): RiseTime, PeakTime, Overshoot, SettlingTime.

4

Compare

Tabulate against theory and compute % error.

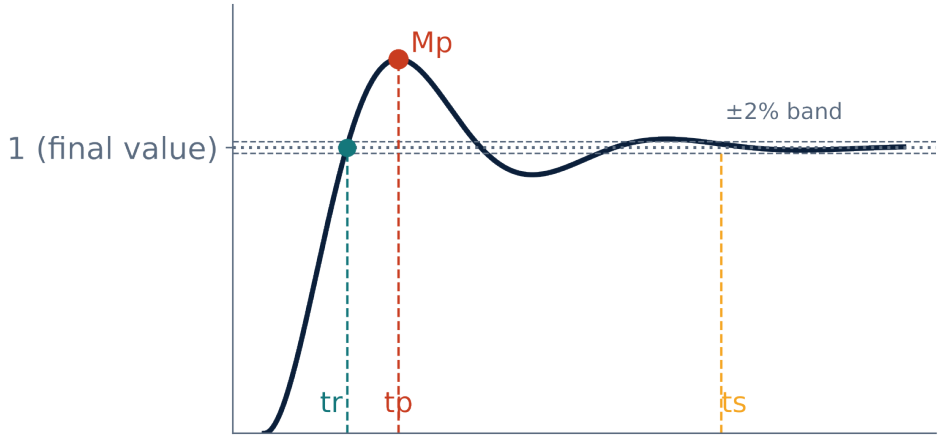
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MATLAB program

```
wn = 5; zeta = 0.6;
sys = tf(wn^2, [1 2*zeta*wn wn^2]);
S = stepinfo(sys, 'SettlingTimeThreshold', 0.02)
wd = wn*sqrt(1-zeta^2);  phi = acos(zeta);
tr = (pi-phi)/wd;  tp = pi/wd;
Mp = 100*exp(-pi*zeta/sqrt(1-zeta^2));  ts = 4/(zeta*wn);
fprintf('tp = %.3f s, Mp = %.2f %%, ts = %.3f s\n', tp, Mp, ts)
step(sys); grid on
```

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Annotated step response



- Mark tr, tp, Mp and ts on the plot (right-click → Characteristics).
- The ±2 % band defines ts.

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Observation table

Parameter	Theory	MATLAB	% error
Peak time tp (s)	0.785		
Peak overshoot Mp (%)	9.48		
Settling time ts (s)	1.33		
Rise time tr (s)	0.554		

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Key takeaways

- stepinfo returns rise, peak, overshoot and settling values
- tp and Mp match theory closely; ts is approximate
- Mp depends only on ζ
- Specifications are read from the step response

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Practice questions

Q1. Find the specifications for ωn = 10, ζ = 0.4.

Q2. Why does MATLAB's rise time differ from the textbook value?

Q3. Write the stepinfo command with a 5 % band.

Q4. How do ts and Mp change if ωn doubles for fixed ζ?